

Shuo HUANG (she/her)

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Research Interests

Theory of Deep Learning: generalization analysis of deep neural networks, operator learning, functional data analysis
Approximation Theory: spherical harmonics, neural networks
Statistical Learning Theory: kernel methods, regression, classification, pairwise learning, etc.

Education

- **City University of Hong Kong, Hong Kong** 2019 – 2023
PhD in Mathematics
Supervisors: [Prof. Han Feng](#) & [Prof. Ding-Xuan Zhou](#)
Thesis Title: [Mathematical Analysis of Deep Neural Networks](#)
- **Hohai University, China** 2015 – 2019
BSc in Applied Mathematics
GPA: 4.91/5.00 (Ranked 1st out of 91)

Work Experience

- **Istituto Italiano di Tecnologia, Genova, Italy** 2024 – Present
Postdoctoral Scholar
Mentor: [Prof. Lorenzo Rosasco](#)
- **Massachusetts Institute of Technology, MA, USA** Sep. – Oct. 2025
Visited [Prof. Tomaso Poggio](#)'s group

Publications and Preprints (* alphabetical order, † corresponding author)

1. Jianfei Li, **Shuo Huang**[†], Han Feng, Ding-Xuan Zhou, Gitta Kutyniok. Sparse-Aware Neural Networks for Nonlinear Functionals: Mitigating the Exponential Dependence on Dimension. *arXiv preprint*, arXiv:2604.06774, 2026.
2. Zhenyu Yang, **Shuo Huang**[†], Han Feng, Ding-Xuan Zhou. Spherical Analysis of Learning Nonlinear Functionals. *Constructive Approximation*, 2026, 1-29.
3. Junyu Zhou, **Shuo Huang**, Han Feng, Puyu Wang, Ding-Xuan Zhou. Fine-grained analysis of non-parametric estimation for pairwise learning. *IEEE Transactions on Neural Networks and Learning Systems*, 2026, 1-12.
4. **Shuo Huang**, Hippolyte Labarrière, Ernesto De Vito, Tomaso Poggio, Lorenzo Rosasco. Learning Multi-Index Models with Hyper-Kernel Ridge Regression. *arXiv preprint*, arXiv:2510.02532, 2025.
5. Franceso Insulla, **Shuo Huang**, Lorenzo Rosasco. Towards a Learning Theory of Representation Alignment. *International Conference on Learning Representations (ICLR)*, 2025.
6. Junyu Zhou, **Shuo Huang**, Han Feng, Ding-Xuan Zhou. Optimal Estimates for Pairwise Learning with Deep ReLU Networks. *arXiv preprint*, arXiv:2305.19640, 2023.
7. **Shuo Huang**, Junyu Zhou, Han Feng, Ding-Xuan Zhou. Generalization Analysis of Pairwise Learning for Ranking with Deep Neural Networks. *Neural Computation*, 35:6 (2023), 1135-1158.
8. Han Feng*, **Shuo Huang***, Ding-Xuan Zhou*. Generalization Analysis of CNNs for Classification on Spheres. *IEEE Transactions on Neural Networks and Learning Systems*, 34:9 (2023), 6200-6213.

9. Zhiying Fang*, Han Feng*, **Shuo Huang***, Ding-Xuan Zhou*. Theory of Deep Convolutional Neural Networks II: Spherical Analysis. *Neural Networks*, 131 (2020), 154-162.

Honors & Awards

- Outstanding Academic Performance Award, City University of Hong Kong 2022
(Awarded annually to Top 25% Ph.D. students)
- Outstanding Graduates, Hohai University 2019
(Recognized as the top graduate of the year in the entire college)
- China National Scholarship, China 2017
(Awarded to the best student in the entire college for exceptional academic performance)
- Second Prize in China Undergraduate Mathematical Contest in Modeling, China 2017
(Awarded for achieving second place in a national competition focused on applying mathematical modeling to real-world problems)
- National Outstanding Volunteer, China 2016
(Recognized for organizing fundraising for underprivileged children and volunteer teaching)

Teaching Experience

Department of Computer Science, University of Genova

- Machine Learning (Master course with Lorenzo Rosasco) Fall 2025
(Taught the Sparsity part)

Department of Mathematics, University of Genova

- Machine Learning (Master course with Lorenzo Rosasco) Fall 2024
(Taught the Decision Tree part)

Department of Mathematics, City University of Hong Kong

- GE1358-MA1501 Coordinate Geometry (Teaching Assistant) Fall 2021-2022
(Led weekly tutorial sessions for 60 students)
- MA2001 Multi-variable Calculus & Linear Algebra (Teaching Assistant) Spring 2020-2022
(Led weekly tutorial sessions, invigilating and grading for 30-90 students)
- MA1201 Calculus & Basic Linear Algebra II (Teaching Assistant) Fall 2020-2021
(Led weekly tutorial sessions, invigilating and grading for 120 students)
- MA4546 Introduction to Stochastic Processes (Teaching Assistant) Spring 2019-2020
(Assisted in grading assignments and exams for 90 students)

Invited Talks and Conferences

- Seminar Talk, TU Delft, Netherlands 12 - 15 Jan. 2026
Statistical Learning of NNs from Structure to Generalization
- Online Seminar Talk, Northwestern University, USA 21 Nov. 2025
Learning Multi-Index Models with Hyper-Kernel Ridge Regression
- Seminar Talk, University of Messina, Italy 6 - 8 Oct. 2025
Learning Multi-Index Models with Hyper-Kernel Ridge Regression
- Group Talk, CBMM, MIT, USA 29 Sep. 2025
Learning Multi-Index Models with Hyper-Kernel Ridge Regression
- ICLR (poster) , Singapore 24 - 28 Apr. 2025
Towards a Learning Theory of Representation Learning

- MFO Workshop-Overparametrization, Regularization, Identifiability and Uncertainty in Machine Learning, Oberwolfach, Germany 26 - 31 Jan. 2025
Beyond the Curse of Dimensionality with Hyper-Kernel Ridge Regression.
- Seminar Talk, Fudan University, China 15 Jun. 2023
Generalization Analysis of CNNs for Classification.
- International Conference on Applied Mathematics, Hong Kong 30 May - 03 Jun. 2023
Generalization Analysis of Pairwise Learning for Ranking with Deep Neural Networks.
- International Conference on Multivariate Approximation, Germany 13 - 17 Mar. 2023
Approximation Ability of Deep Convolutional Neural Networks.

Organizations

- [Seminars](#) at Machine Learning Genova Center (MaLGa) 2025

Supervision

- Lorenzo Fiorito (Ph.D student at Genova) Nov. 2024 - Present
- Even Martel (Master student at ENS de Lyon) Apr. - Jul. 2025
- Francesco Insulla (Master student at Stanford University) Jun. - Jul. 2024

Professional Service

Reviewer for Journals: Neural Networks, IEEE Transactions on Neural Networks and Learning Systems, Journal of Machine Learning Research, IEEE Transactions on Information Theory, Analysis and Applications, etc.

Reviewer for Conferences: ICLR, ICML, AAAI, etc.